

Multimodal Inputs

Multimodal inputs refer to AI systems that can understand and process multiple types of data such as text, images, audio, and documents together. Instead of relying only on text, modern AI models are designed to interpret different input formats and combine them to produce more meaningful results. For example, vision-capable models can take both an image and a text prompt as input. A user might upload a picture and ask the model to describe it, detect objects, extract information from it, or answer questions about what is visible in the image. Another important multimodal capability is audio processing. Models such as **Whisper** convert spoken language into text through an audio transcription pipeline, which is widely used in voice assistants, meeting transcription tools, and accessibility applications. Document understanding is another key component of multimodal AI. Many real-world documents such as PDFs, invoices, forms, and reports contain mixed content including text, tables, images, and complex layouts. AI systems must analyze the document structure and extract meaningful information from these unstructured formats. Within this process, **Optical Character Recognition (OCR)** plays an important role. OCR converts text present in images or scanned documents into machine-readable text. Rather than being treated as a standalone task, OCR is usually one stage in a larger **Document AI pipeline** that includes layout detection, table extraction, entity recognition, and semantic understanding of the document.